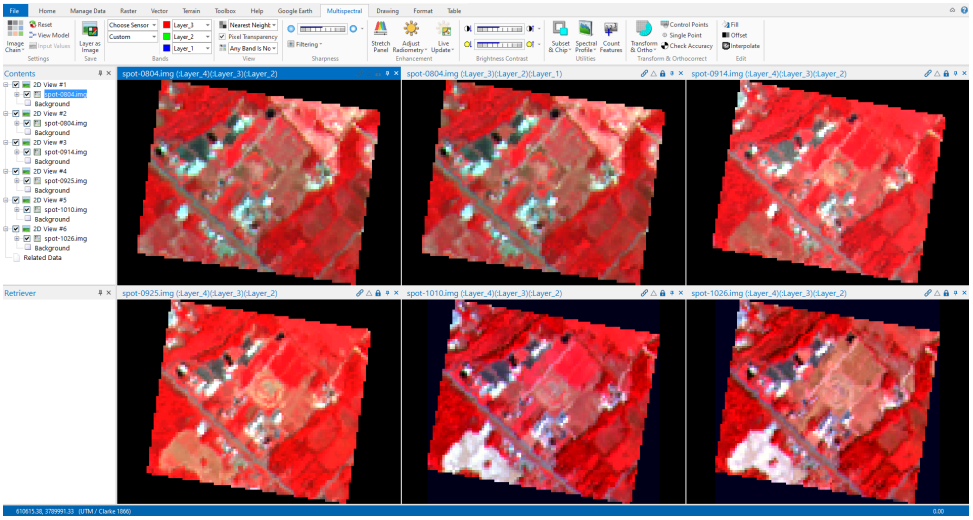
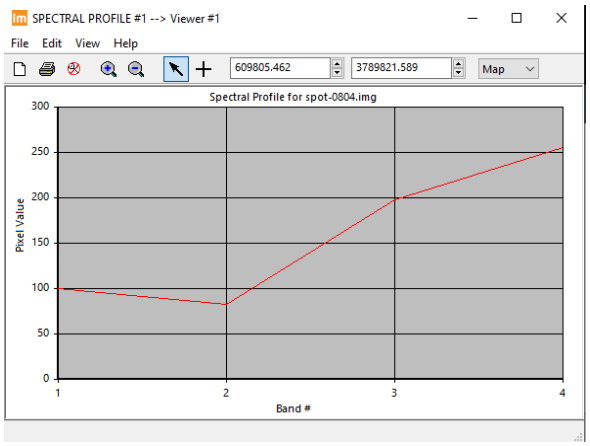
Part A

1.)

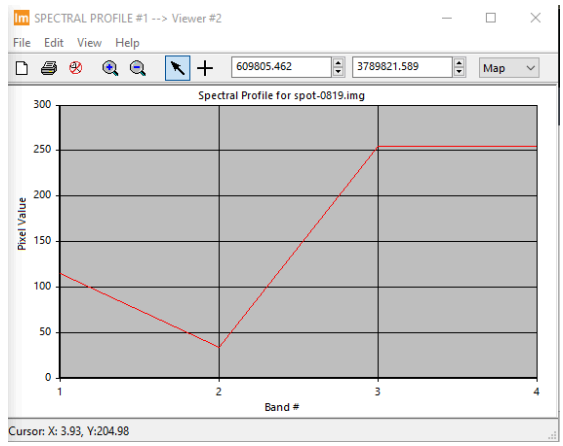


2.) ..

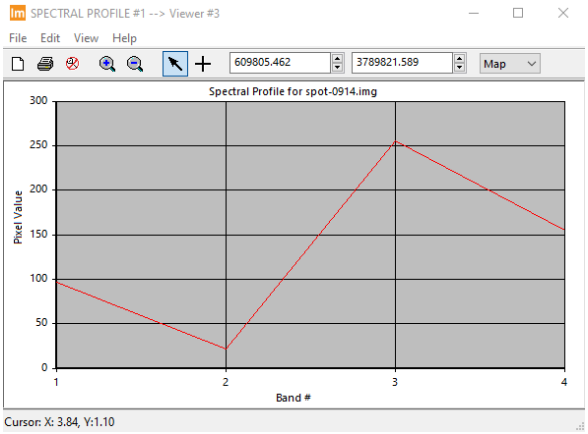
spot-0804: August 4, 1998



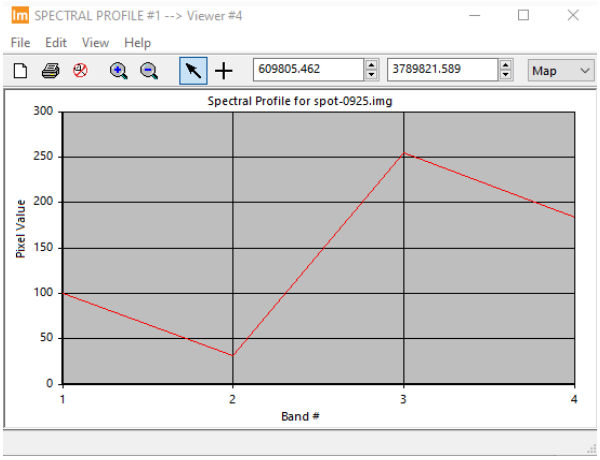
spot-0819: August 19, 1998



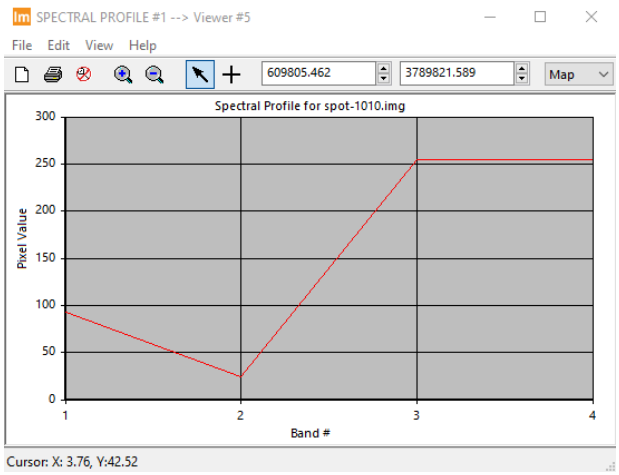
spot-0914: September 14, 1998



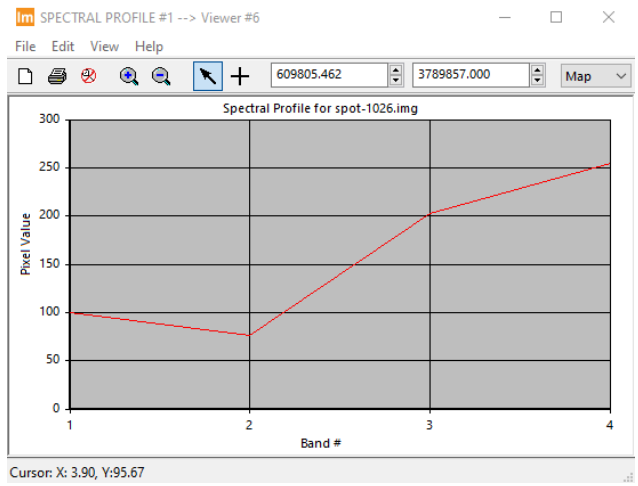
spot-0925: September 25, 1998



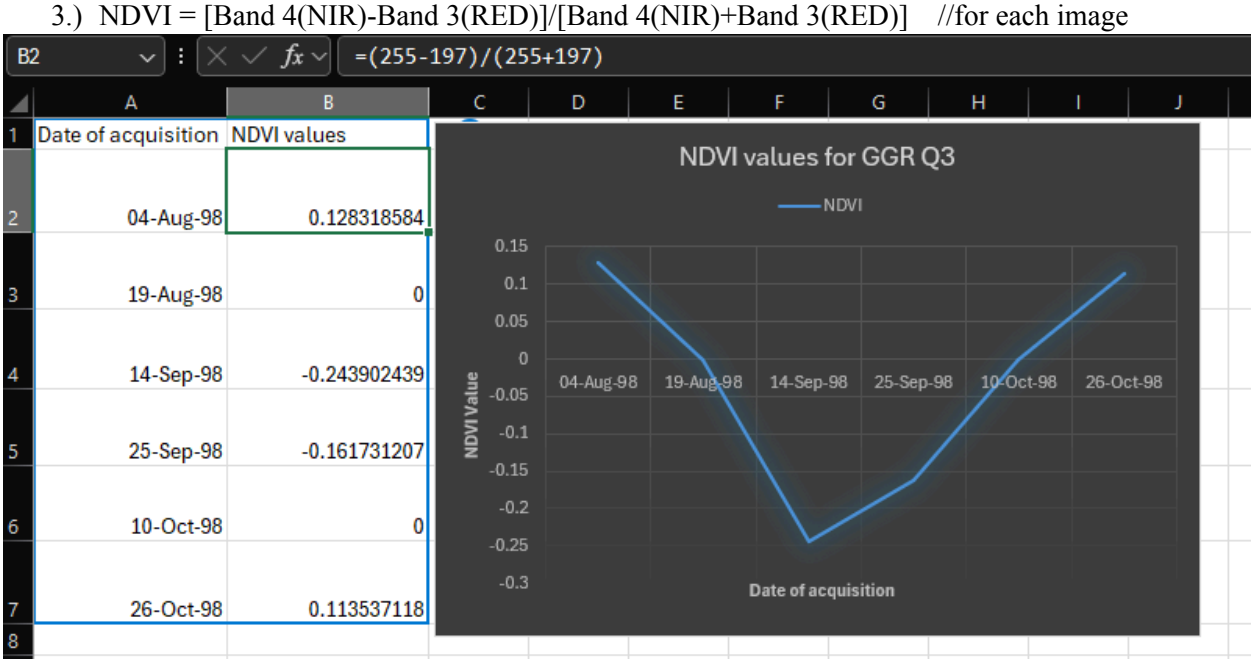
spot-1010: October 10, 1998



spot-1026: October 26, 1998



3.) NDVI = [Band 4(NIR)-Band 3(RED)]/[Band 4(NIR)+Band 3(RED)] //for each image



Date of acquisition	NDVI values
04-Aug-98	0.128318584
19-Aug-98	0
14-Sep-98	-0.243902439
25-Sep-98	-0.161731207
10-Oct-98	0
26-Oct-98	0.113537118

4.) The range of NDVI values are -1 to 1. A **high NDVI** value (closer to 1) indicates healthy, dense vegetation with a strong chlorophyll signal. **Low NDVI** values (closer to 0 or negative) suggest sparse or no vegetation.

The red and near-infrared bands are particularly useful for vegetation indices such as NDVI because the **Red band** measures absorption by chlorophyll, which decreases during senescence.The **NIR band** measures reflectance from cell structures, which remains high in healthy vegetation. The contrast between these two bands makes them ideal for detecting vegetation health.

5.) **5-Aug-98:**Reflectance in the red band (B3) is relatively high due to limited chlorophyll absorption, while the near-infrared band (B4) shows moderate reflectance as some photosynthetic activity occurs.

19-Aug-98: An NDVI of 0 means there is little or no vegetation growth. This could be due to factors like weather, poor nutrients, or other stresses. The red and near-infrared reflectance are similar, showing minimal vegetation activity**.**

14-Sep-98:NDVI values decline as the plants lose chlorophyll and healthy structure. The reflectance characteristics shift, with increasing red reflectance and decreasing near-infrared reflectance.

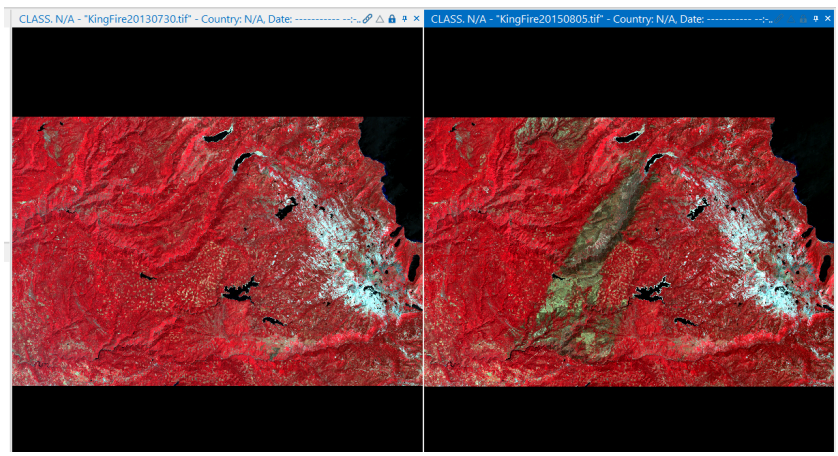
25-Sep-98:The negative NDVI continues to reflect ongoing senescence, where much of the crop's canopy has either died or fallen, further reducing near-infrared reflectance. Exposed soil or stubble may dominate the reflectance signal, and the field is transitioning into a largely non-vegetative state.

10-Oct-98:The NDVI value returning to 0 indicates minimal vegetation cover. By this point, the soybean crop may have been harvested or has completely senesced, leaving bare soil or crop residue.

26-Oct-98:NDVI values rise slightly, possibly due to regrowth or cover crops.Healthy, photosynthetically active vegetation strongly absorbs red light (low red reflectance) and reflects high near-infrared, leading to higher NDVI values.

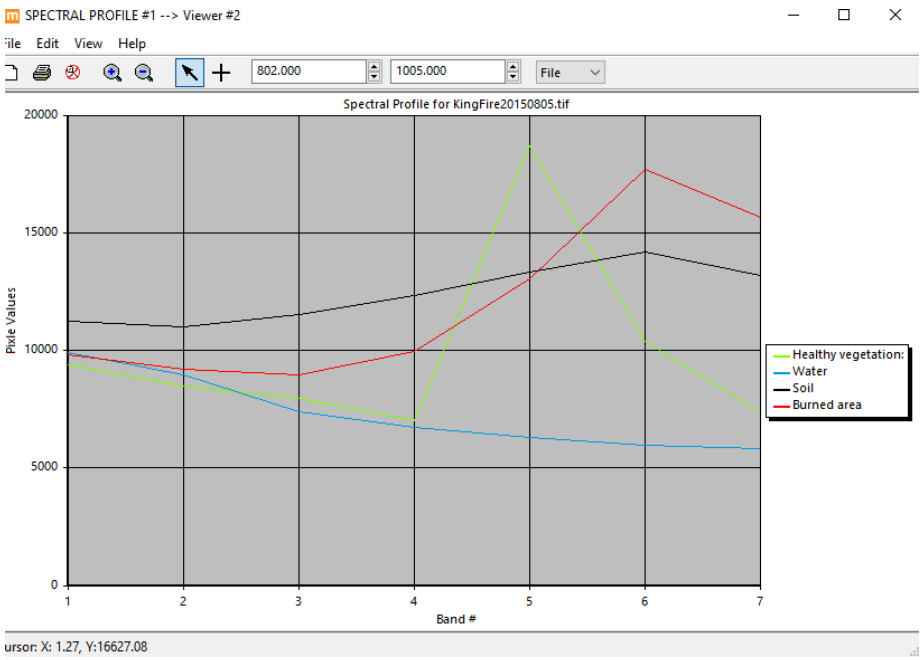
Part B

6.)



True Color Composite (TCC): Burned areas appear dark brown due to loss of vegetation and exposed soil.
False Color Composite (FCC): Burned areas will have minimal NIR reflectance of vegetation(close to none) and replaced with a higher reflectance in visible bands.

This is because burned vegetation reflects more visible light and less NIR due to the loss of chlorophyll and vegetation structure.

7.) 

8.) The digital number values of **SWIR** (Bands 6, 7)bands are generally higher in the burned area than the healthy forest.This is because the **Red Band(ranging 0.64 - 0.67 µm**) detects vegetation health and fire damage due to changes in chlorophyll. The **SWIR Band 6 (1.57-1.65 µm)** and **SWIR Band 7 (2.11 - 2.29 µm)** are sensitive to moisture content and structural changes, making it ideal for identifying burned areas and soil properties.

9.) UTM coordinate:10s 714228 4315261 with …

NBR=(Band 5 - Band 7)/(Band 5+ Band 7) = (9696 - 15838)/(9696+ 15838)≈ -0.24

SR= Band 5/ Band 4 = 9696/12193 ≈ 0.79

A low NBR value indicates loss of vegetation, with low NIR reflectance (Band 5) and high SWIR reflectance (Band 7), typical of burned areas. Lower SR values also means less vegetation reflects in the NIR band due to destruction of the leaf structure.

10.)NBR_PreFire

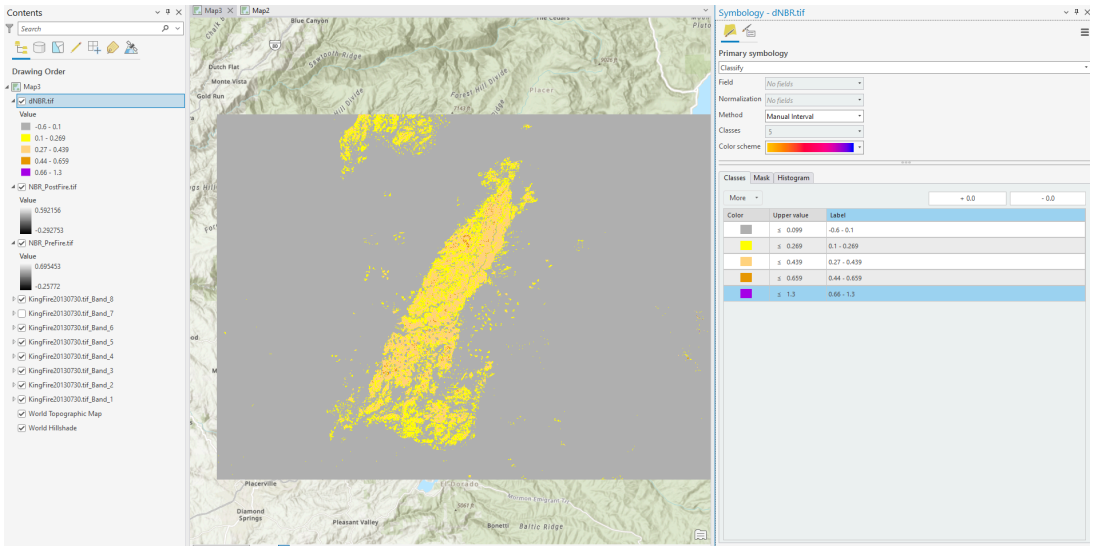
NBR_PostFire

11.) **Pre-fire:** High NBR values indicate healthy vegetation with strong NIR and low SWIR reflectance.

Post-fire: Lower NBR values indicate burned areas with reduced vegetation and exposed soil.

This is because fire removes vegetation, reduces NIR reflectance, and increases SWIR reflectance due to structural changes and moisture loss.

12.)Classified burn severity image



13.) Most areas in the center are of moderate severity. The outer edges are low severity. When zooming in in there are small patch that are high severity

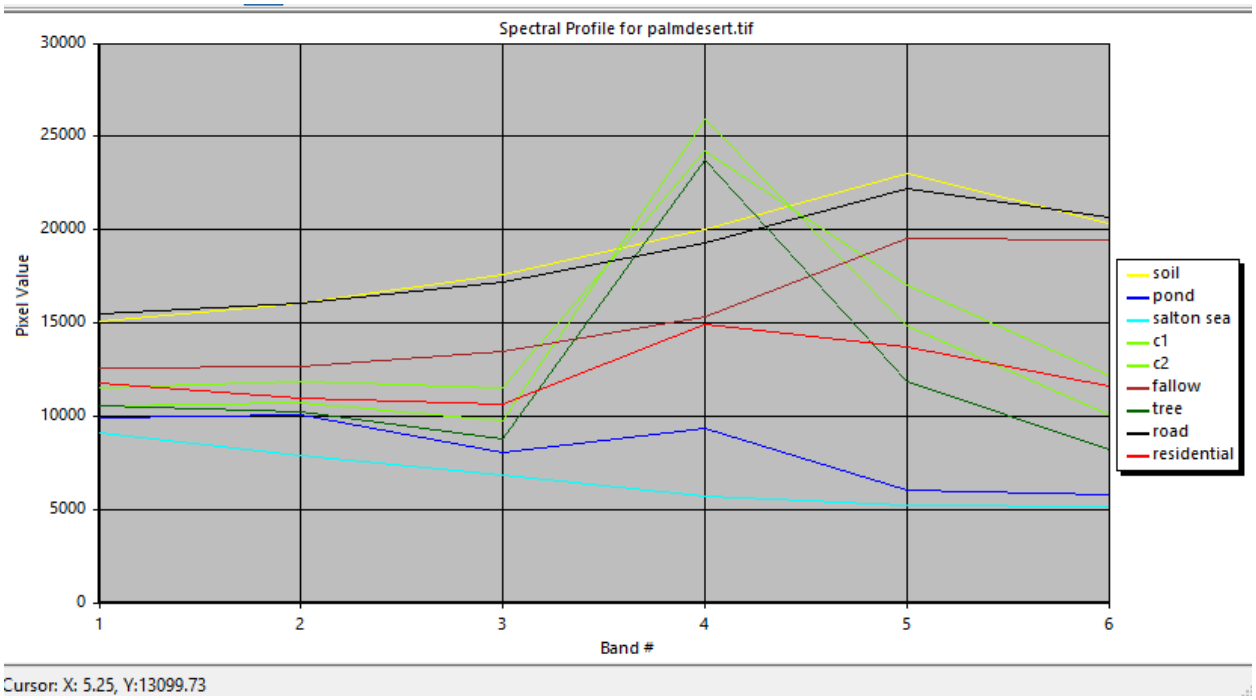
14.)Unburned areas within the fire perimeter are often associated with water bodies or terrain types that lack sufficient fuel for fire.The main environmental difference between burned and unburned areas in 2020 is the recovery of vegetation. Unburned areas maintain dense vegetation, while burned areas exhibit slower regrowth or bare ground, reflecting the long-term impact of the fire.

Part C

15.).

Land Cover	**Position(x,y,meter) Zone 11**	**Band 1**	**Band 2**	**Band 3**	**Band 4**	**Band 5**	**Band 6**
bare soil	603443,3706802	15094	16027	17571	19859	22981	20351
pond	584103,3708393	10141	11125	9454	6791	6160	6027
Salton Sea	585778,3705080	9 152	7949	6885	5714	5232	5169
cropland 1 (with crop)	592630,3713410	10088	10708	9688	26968	14807	9484
cropland 2 (different crop)	592536,3713820	10497	11285	11218	27168	17137	10797
fallow	590181,3712705	13395	14120	15553	18856	23475	21950
trees	589313,3709462	10198	9862	9199	24375	12509	8634
road	587194,3713167	15206	16035	17192	19801	23578	22145
residential	584818,3708902	13421	12686	12981	17427	17429	14784

16.)



17.) **Bare Soil**: Moderate reflectance across bands, increasing in SWIR (Bands 5 and 6) due to soil composition and moisture.
Pond/Salton Sea: Very low reflectance in all bands, except slightly higher in Blue (Band 1).
Cropland 1&2: High NIR (Band 4) reflectance from vegetation and low Red (Band 3) reflectance due to chlorophyll absorption.
Fallow: Similar to bare soil but with slightly lower SWIR reflectance due to less moisture.
Trees: High NIR (Band 4) reflectance due to healthy vegetation, with low Red (Band 4) reflectance.
Residential: Mixed reflectance across bands from asphalt, concrete, and vegetation.